Course Notes

Lecture 12 — Message-Oriented Middleware: MQTT and AMQP

Communication Technologies for the Internet of Things

Academic year 2026–2027

Catarina Silva

ESTGA, University of Aveiro

Text to support independent study

Contents

How to Use These Notes **2**

1 Commentary on the Slides **2**

2. Agenda (1/2) . . . 2
3. Agenda (2/2) . . . 3
4. Motivation: Scaling Requires Flexibility (1/2) . . . 4
5. Motivation: Scaling Requires Flexibility (2/2) . . . 5
6. Direct or Through a Broker . . . 6
7. What Message-Oriented Middleware Is (1/2) . . . 7
8. What Message-Oriented Middleware Is (2/2) . . . 8
9. The Three Decouplings . . . 9
10. Distribution Models (1/2) . . . 10
11. Distribution Models (2/2) . . . 11
12. Queues and Publish-Subscribe . . . 12
13. AMQP: Advanced Message Queuing Protocol (1/2) . . . 13
14. AMQP: Advanced Message Queuing Protocol (2/2) . . . 14
15. AMQP: How It Works (1/2) . . . 15
16. AMQP: How It Works (2/2) . . . 16
17. From Producer to Consumer in AMQP . . . 17
18. AMQP: Advanced Features (1/2) . . . 18
19. AMQP: Advanced Features (2/2) . . . 19
20. AMQP: Benefits of Being an Open Standard (1/2) . . . 20
21. AMQP: Benefits of Being an Open Standard (2/2) . . . 21
22. AMQP: Implementations (1/2) . . . 22
23. AMQP: Implementations (2/2) . . . 23
24. MQTT: Message Queue Telemetry Transport (1/2) . . . 24
25. MQTT: Message Queue Telemetry Transport (2/2) . . . 25
26. MQTT: Quality-of-Service Levels (1/2) . . . 26
27. MQTT: Quality-of-Service Levels (2/2) . . . 27
28. The Exchanges of Each QoS Level . . . 28
29. MQTT: Retained Messages and Topics (1/2) . . . 29
30. MQTT: Retained Messages and Topics (2/2) . . . 30
31. Topics and Wildcards . . . 31
32. MQTT: Adoption (1/2) . . . 32
33. MQTT: Adoption (2/2) . . . 33
34. Comparing AMQP and MQTT (1/2) . . . 34
35. Comparing AMQP and MQTT (2/2) . . . 35
36. AMQP and MQTT in a Table . . . 36
37. Which to Choose . . . 37
38. Lecture Summary (1/2) . . . 38
39. Lecture Summary (2/2) . . . 39

Final Check **40**

References **40**

How to Use These Notes

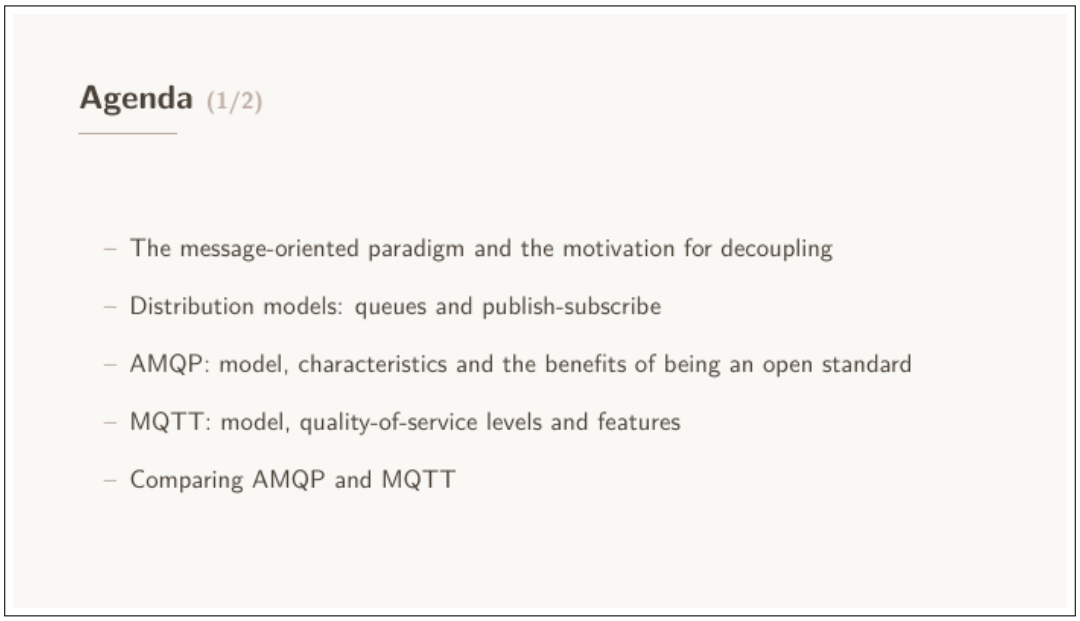
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

1. Commentary on the Slides

Slide 2. Agenda (1/2)



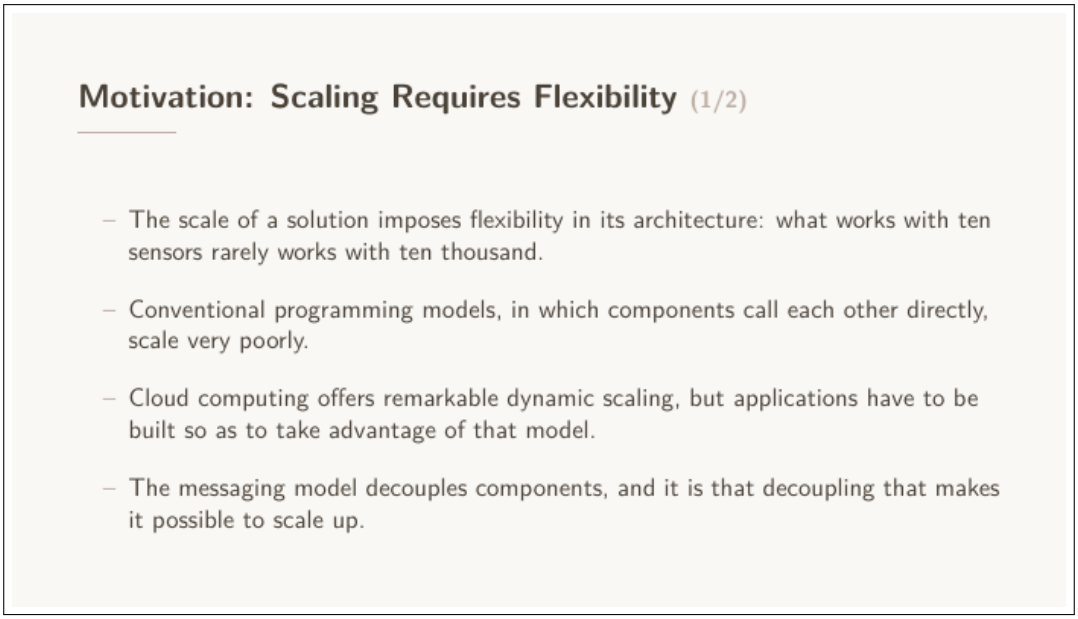
The path through the lecture. The lecture moves from problem to solution. First it justifies decoupling, then it presents the two distribution models, and only then the two protocols, AMQP and MQTT, comparing them at the end.

Slide 3. Agenda (2/2)



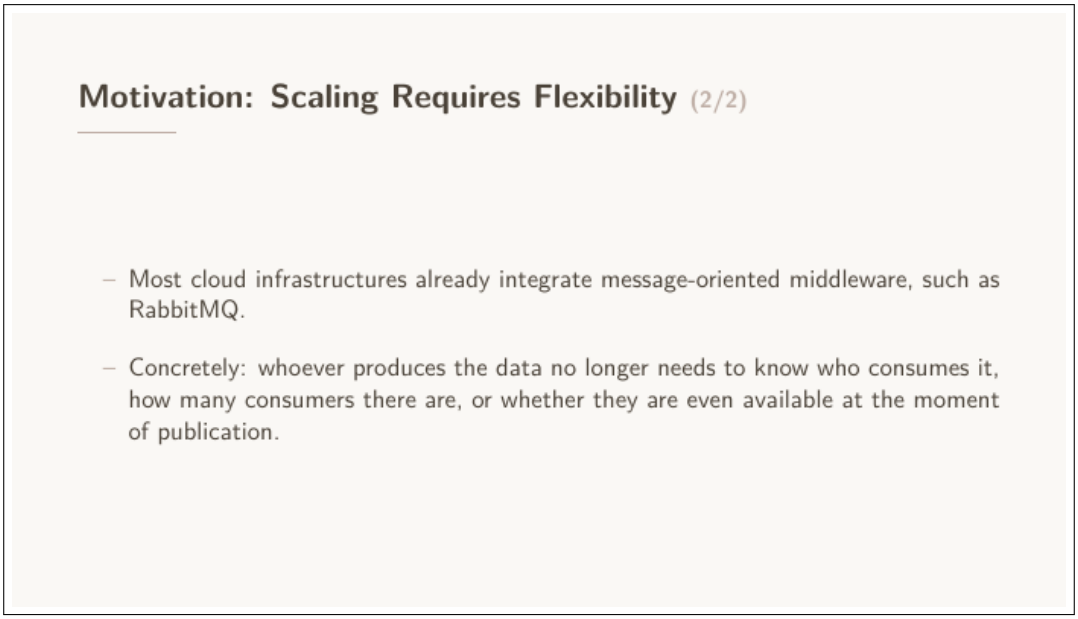
Adoption and criteria. The implementations and adoption show that both protocols are used in production. The selection criteria close the lecture, and are what the student should take to the mini-project.

Slide 4. Motivation: Scaling Requires Flexibility (1/2)



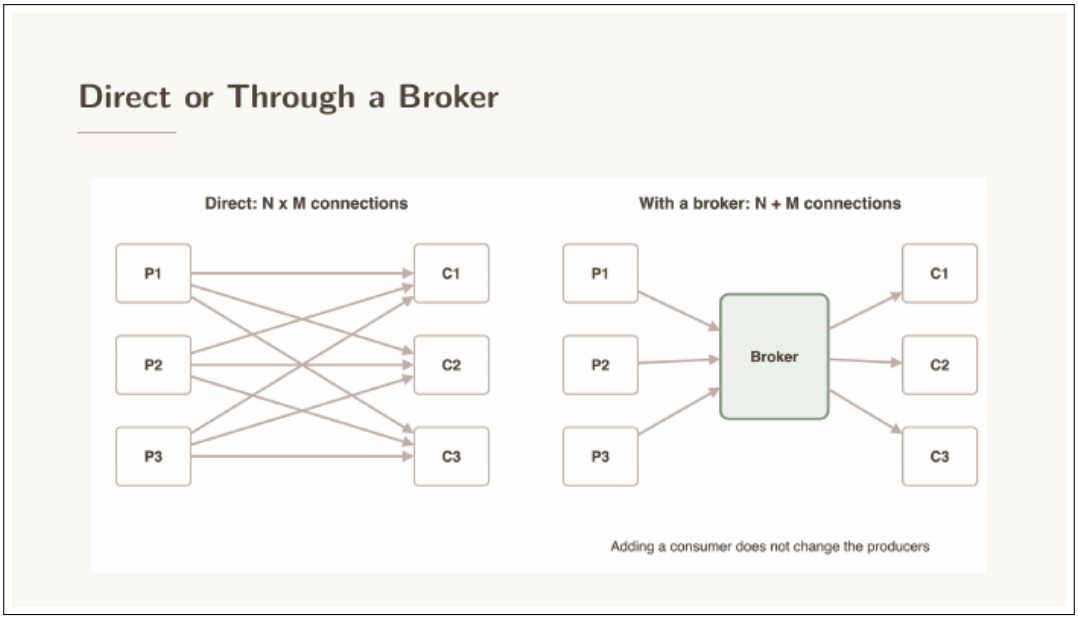
Scaling is decoupling. What works with ten sensors rarely works with ten thousand. When components call each other directly, each new participant forces the others to change. The messaging model cuts that dependency.

Slide 5. Motivation: Scaling Requires Flexibility (2/2)



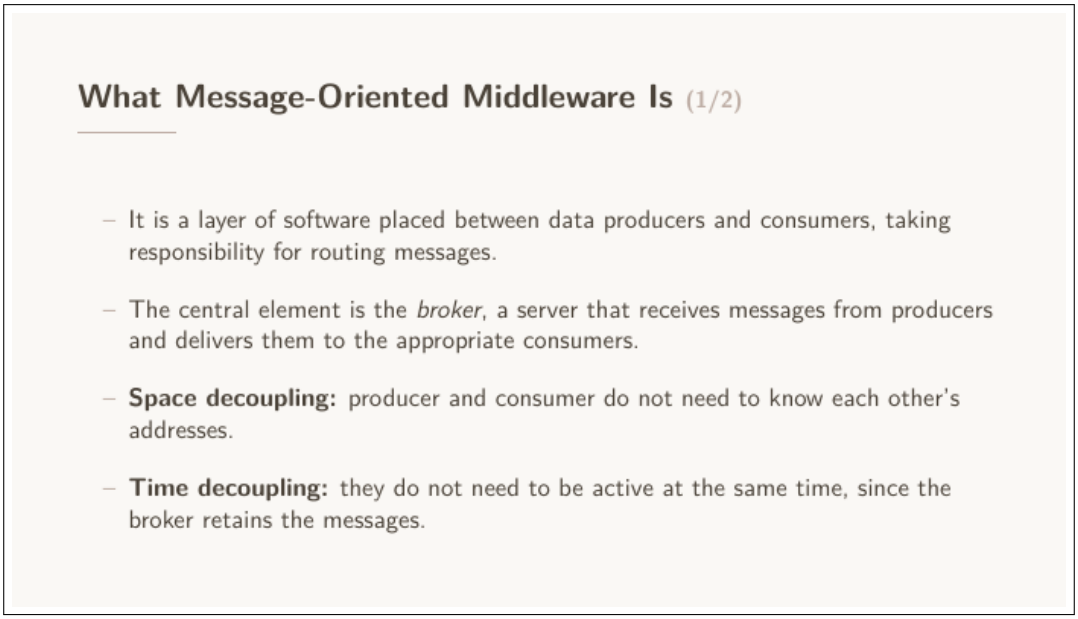
Whoever produces need not know. With a broker, the producer no longer needs to know who consumes, how many there are or whether they are available. It is the property cloud infrastructure exploits, with brokers such as RabbitMQ.

Slide 6. Direct or Through a Broker



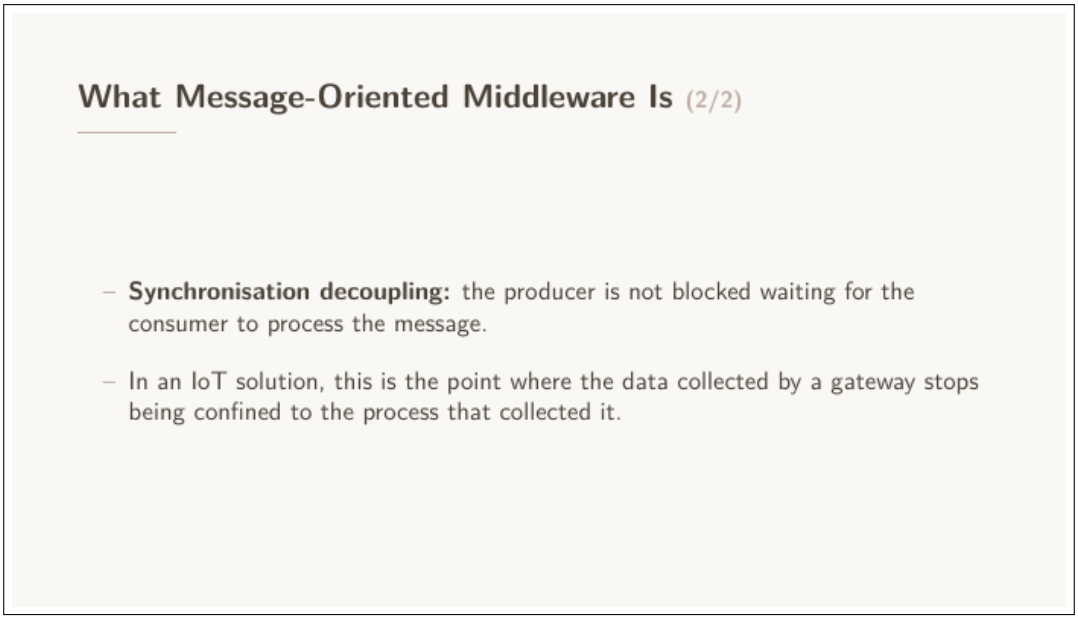
N times M against N plus M. On the left, each producer connects to each consumer: the number of connections is the product. On the right, each connects only to the broker: the number is the sum. Adding a consumer does not change the producers.

Slide 7. What Message-Oriented Middleware Is (1/2)



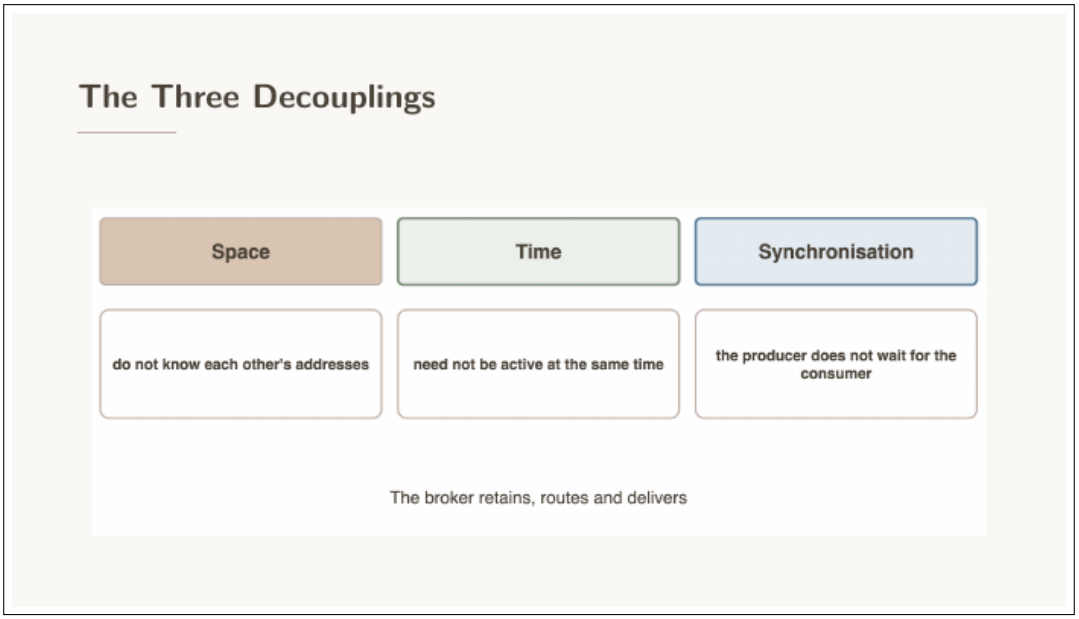
The broker. MOM is a layer between producers and consumers, and the broker is its central element: it receives, retains, routes and delivers. The first two decouplings follow from it.

Slide 8. What Message-Oriented Middleware Is (2/2)



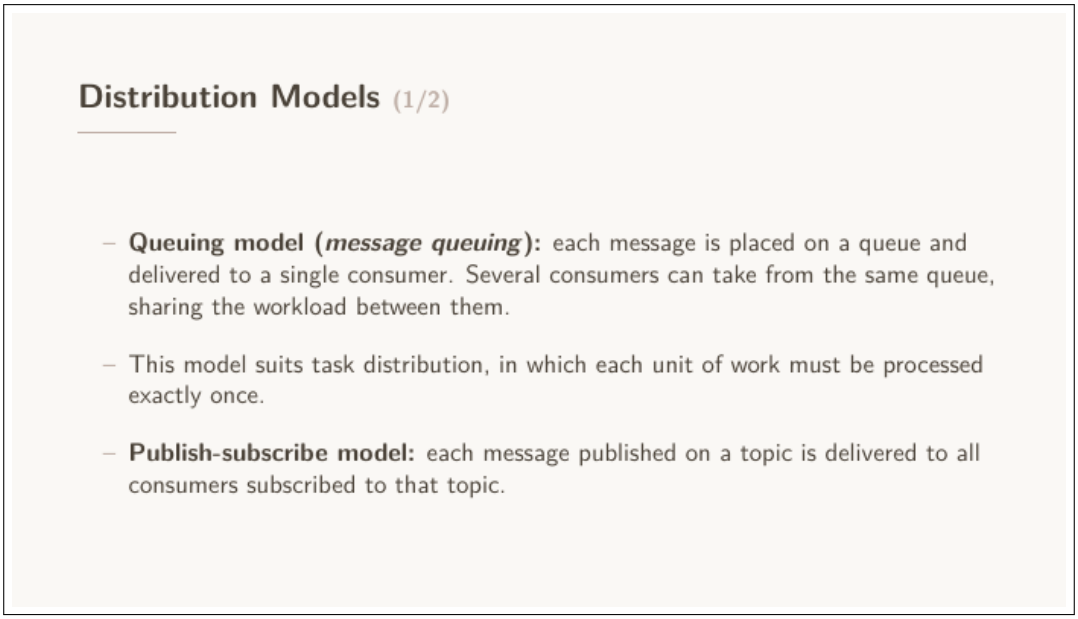
Synchronisation. The third decoupling is synchronisation: the producer publishes and moves on. In IoT, the data collected by a gateway stop being tied to the process that collected them.

Slide 9. The Three Decouplings



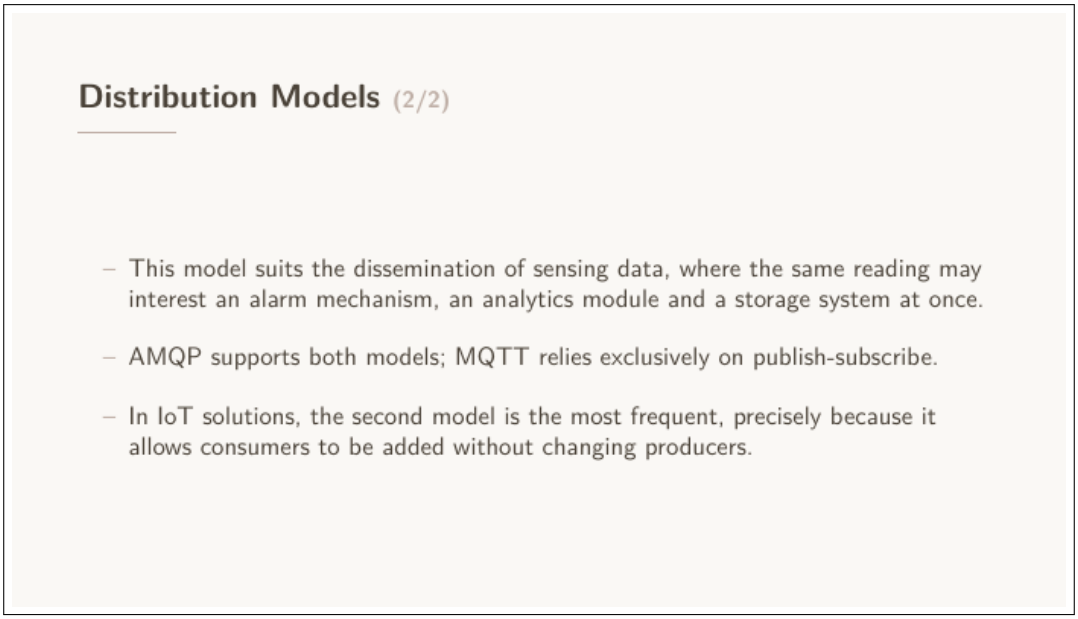
The three decouplings. In space, in time and in synchronisation. Check: can you give an example of each in a sensor scenario?

Slide 10. Distribution Models (1/2)



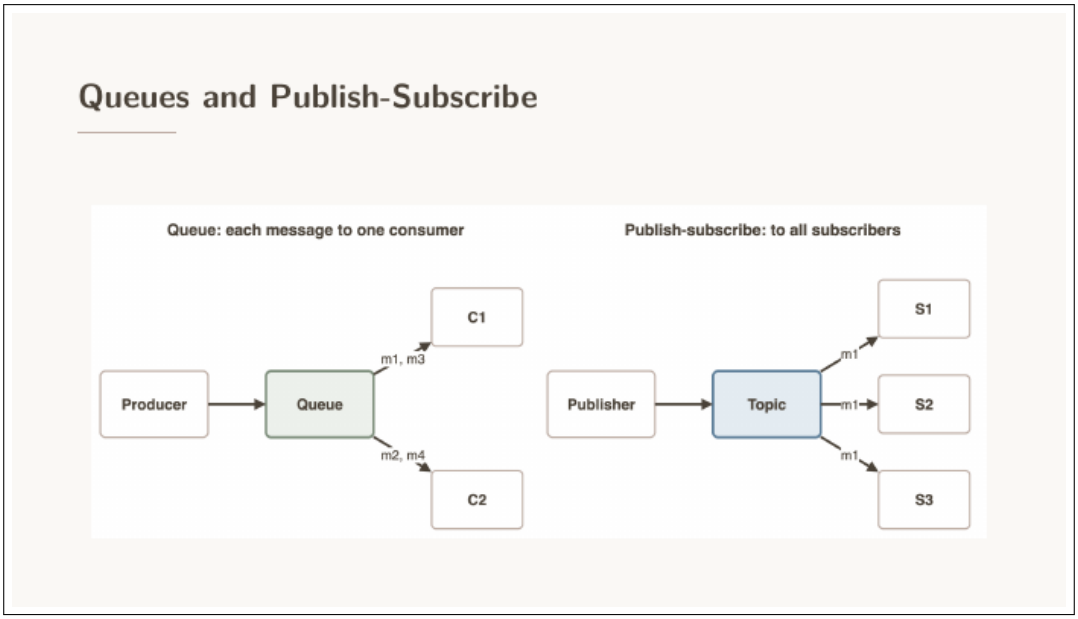
Queue and publish-subscribe. In the queue, each message goes to a single consumer, and several consumers share the work. In publish-subscribe, each message goes to all subscribers of the topic.

Slide 11. Distribution Models (2/2)



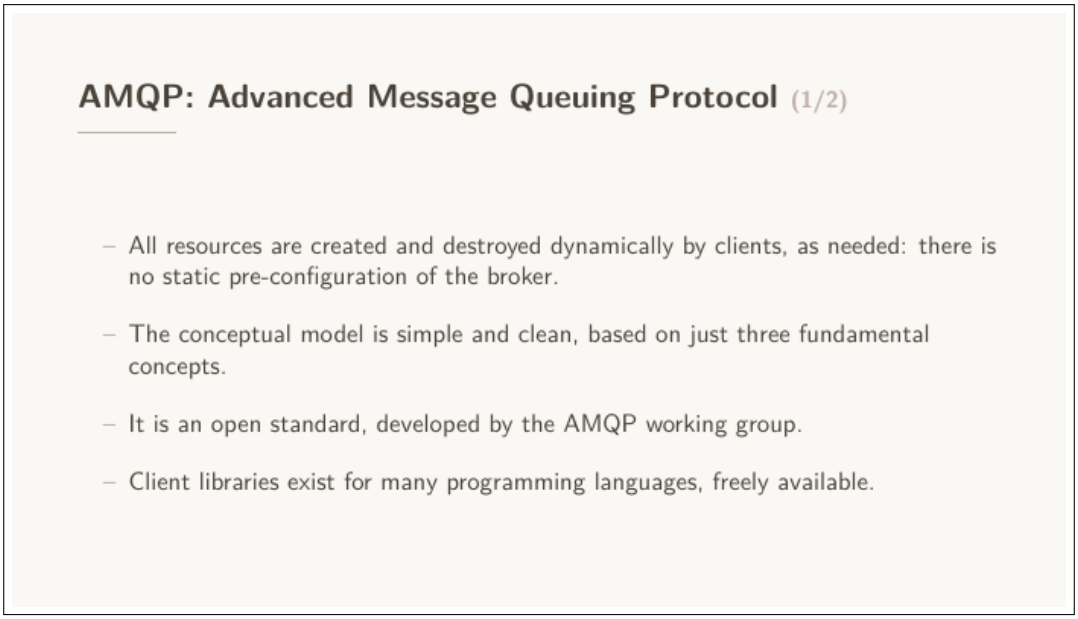
When to use each model. The queue serves to distribute tasks processed exactly once. Publish-subscribe serves to disseminate sensing data to several consumers. In IoT, the second is the most common.

Slide 12. Queues and Publish-Subscribe



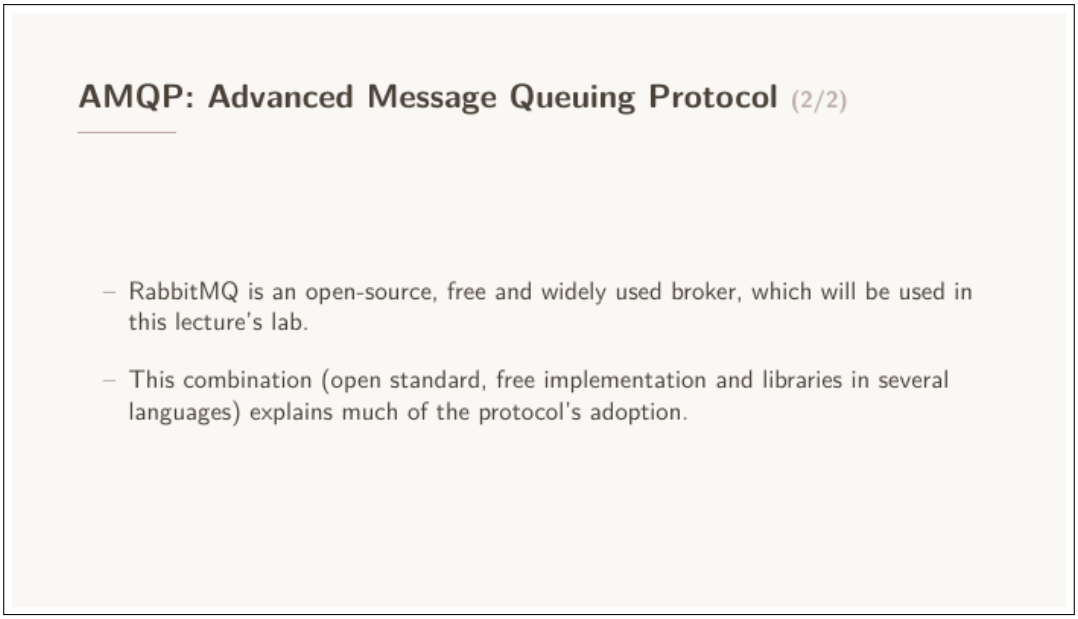
The two models in a picture. On the left, messages m1 to m4 are shared between two consumers of the same queue. On the right, message m1 reaches all subscribers of the topic. It is the difference between sharing and broadcasting.

Slide 13. AMQP: Advanced Message Queuing Protocol (1/2)



AMQP: dynamic and open. In AMQP, resources are created and destroyed by clients, with no pre-configuration of the broker. The model rests on three concepts and is an open standard, with libraries in many languages.

Slide 14. AMQP: Advanced Message Queuing Protocol (2/2)



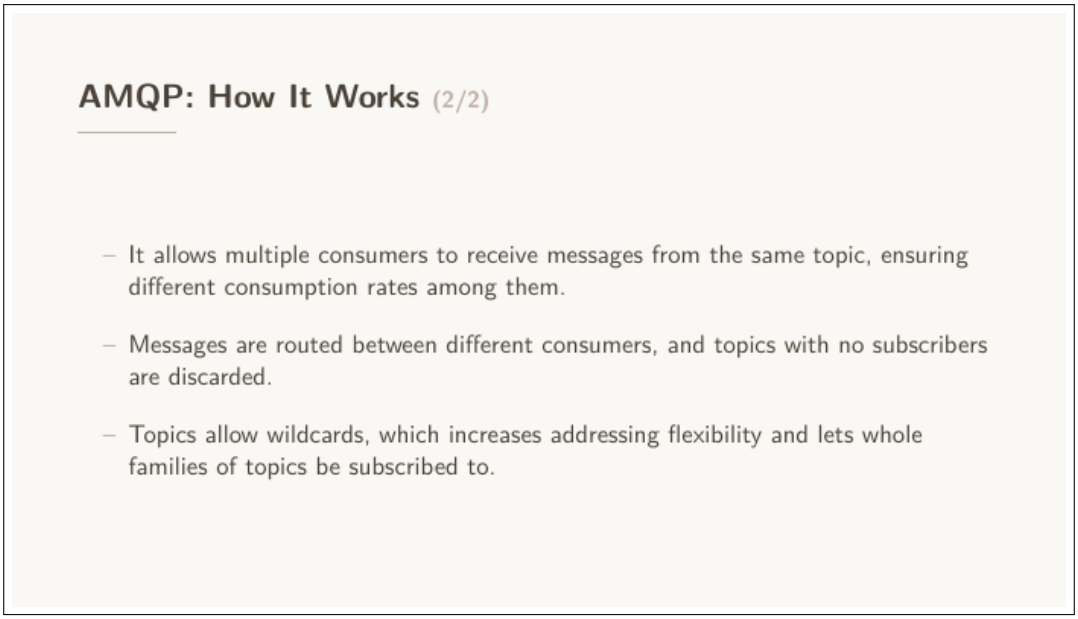
RabbitMQ. RabbitMQ is the most widely used open-source broker and is the one in the lab. The combination of open standard, free implementation and libraries in several languages explains the adoption.

Slide 15. AMQP: How It Works (1/2)



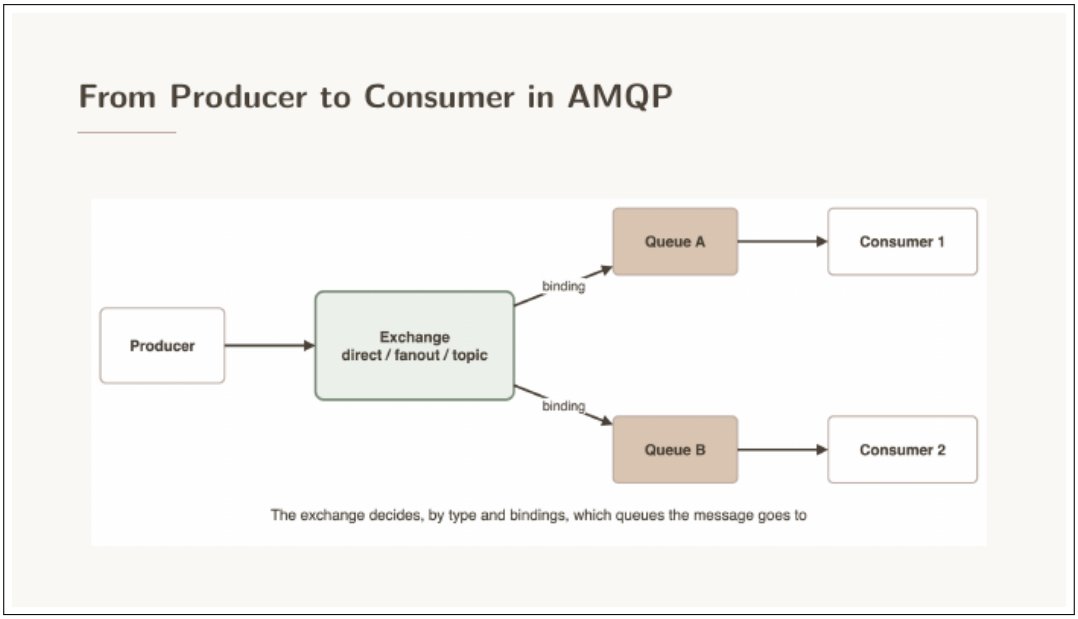
Message, exchange and queue. The producer hands the message to an exchange, and it is the exchange that decides, by type and configured bindings, which queues it goes to. The message waits on the queue until consumed, and order is preserved.

Slide 16. AMQP: How It Works (2/2)



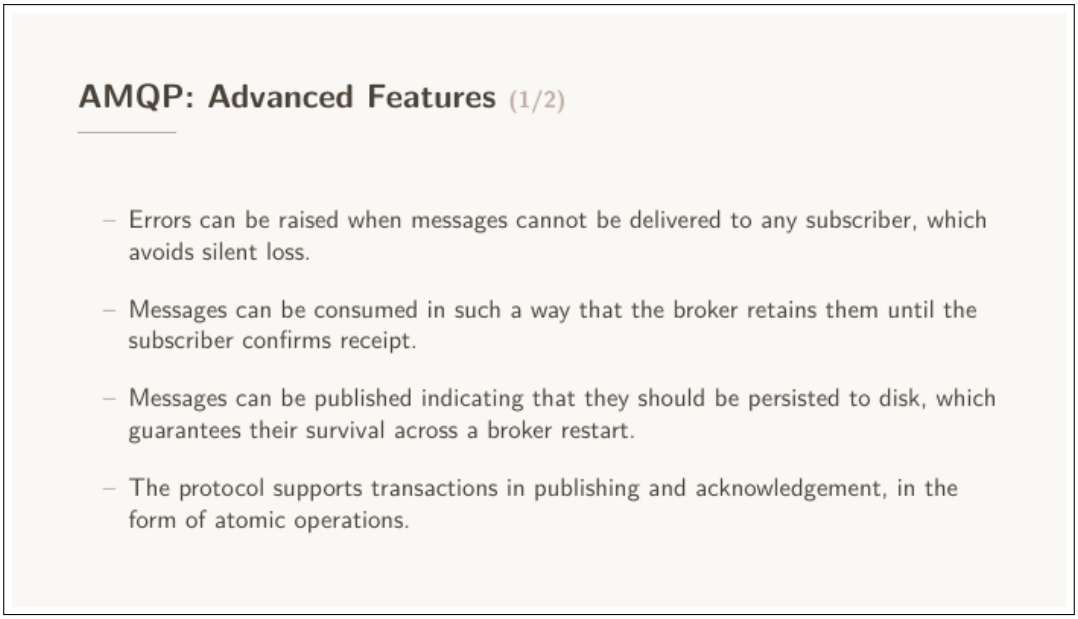
Topics and wildcards. Topics allow wildcards, which makes it possible to subscribe to whole families of topics with a single subscription.

Slide 17. From Producer to Consumer in AMQP



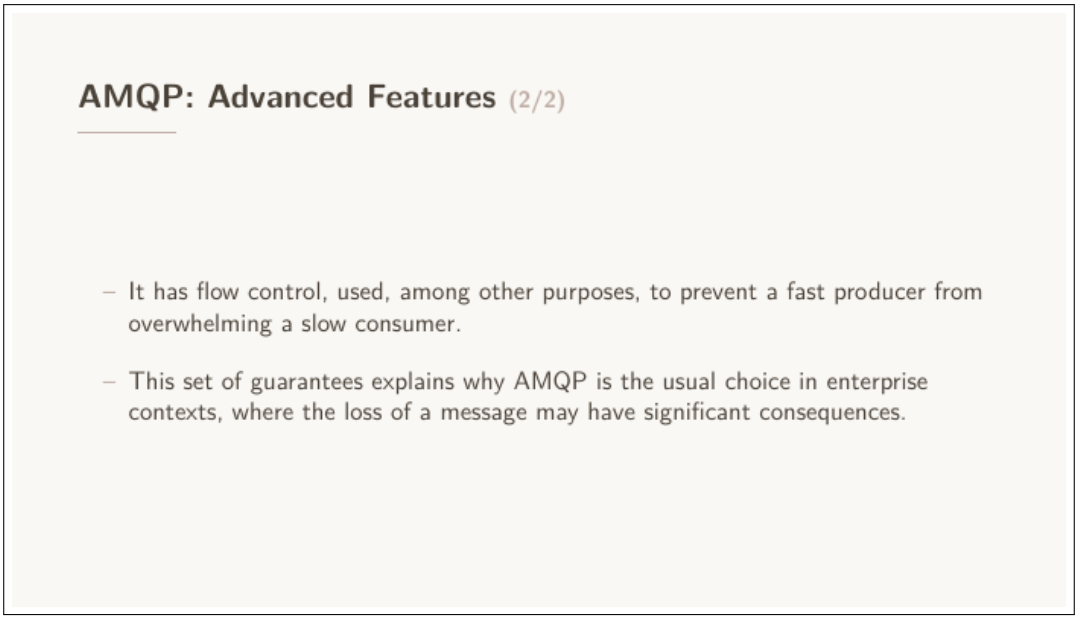
A message's path in AMQP. The figure shows producer, exchange, bindings, queues and consumers. The exchange is the element that routes, and the bindings define to which queues. Check: what changes if the exchange is of the fanout type?

Slide 18. AMQP: Advanced Features (1/2)



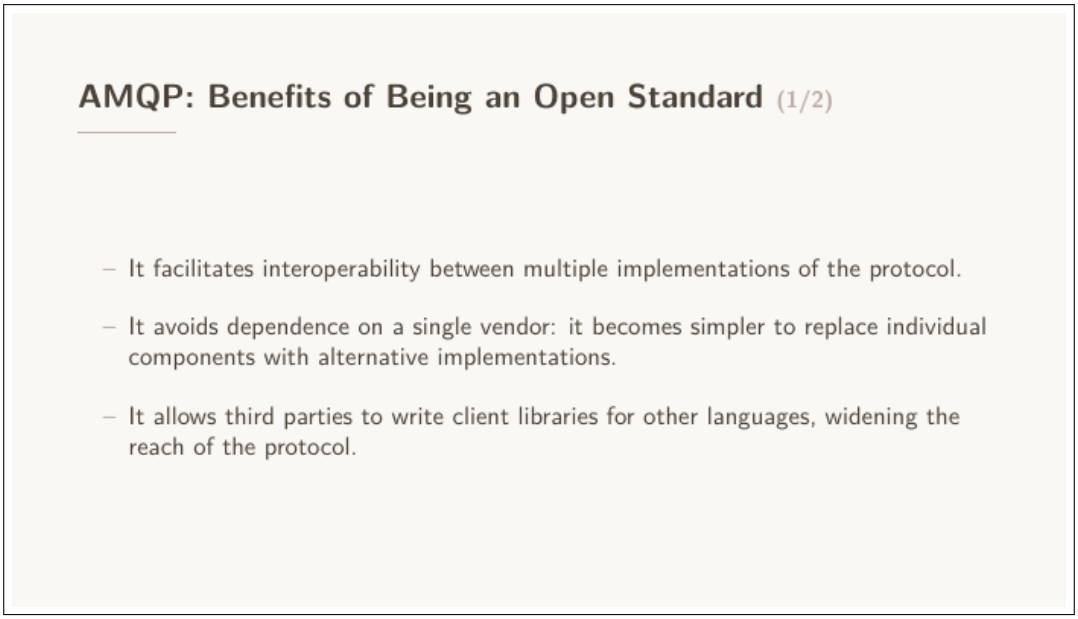
Delivery guarantees. Errors when there is no recipient, retention until acknowledgement, persistence to disk and atomic transactions. Each guarantee costs resources, and is justified in enterprise contexts.

Slide 19. AMQP: Advanced Features (2/2)



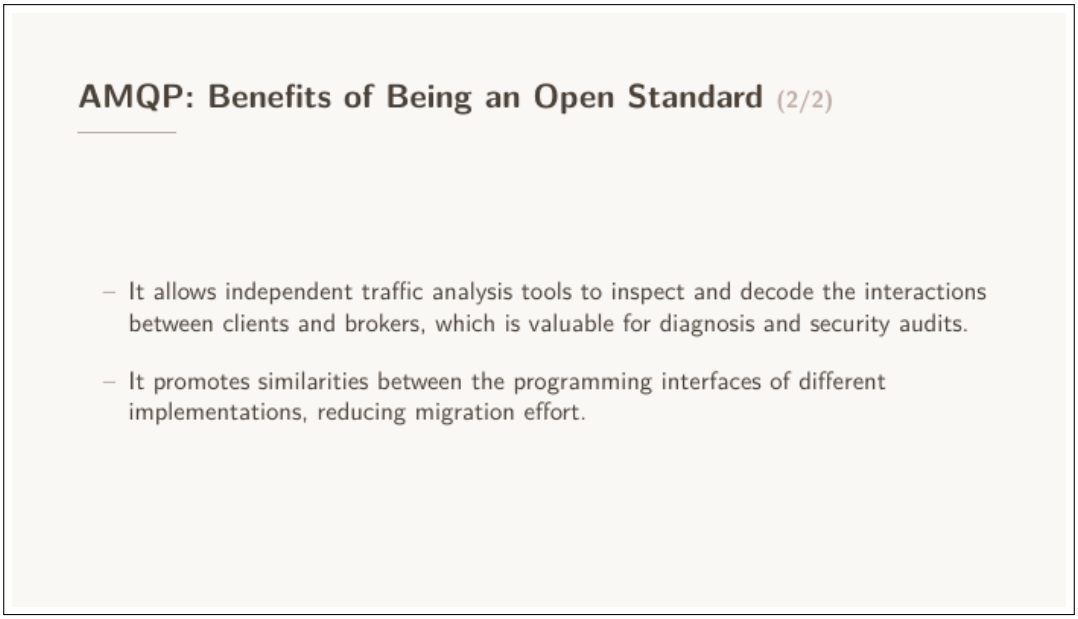
Flow control. Flow control prevents a fast producer from drowning a slow consumer. Together, these guarantees explain AMQP in contexts where losing a message has consequences.

Slide 20. AMQP: Benefits of Being an Open Standard (1/2)



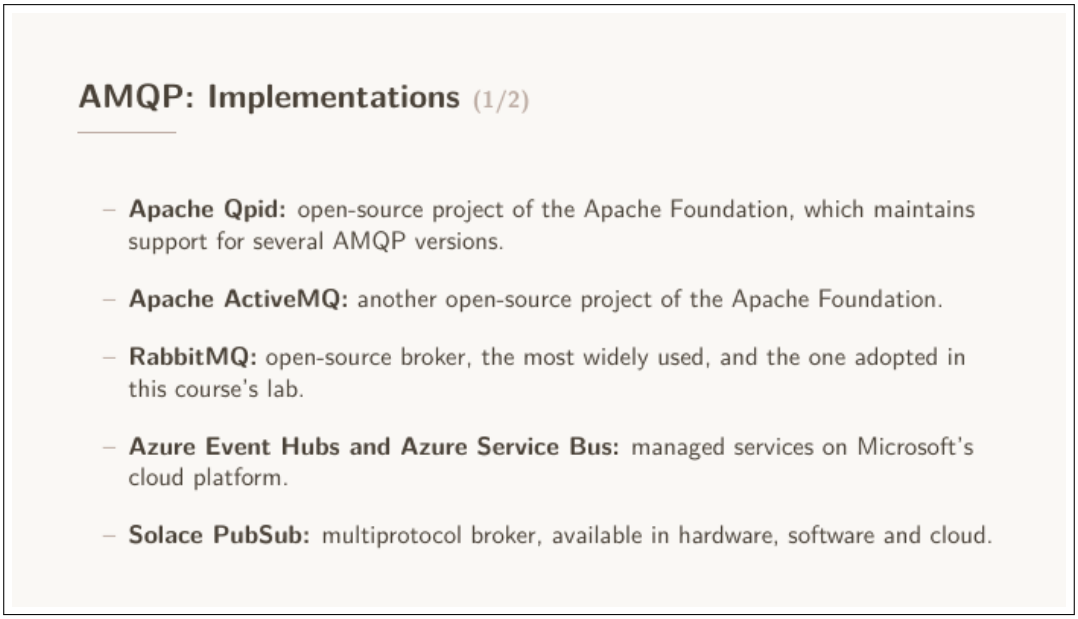
Interoperability. An open standard eases interoperability, avoids depending on a single vendor and lets third parties write client libraries.

Slide 21. AMQP: Benefits of Being an Open Standard (2/2)



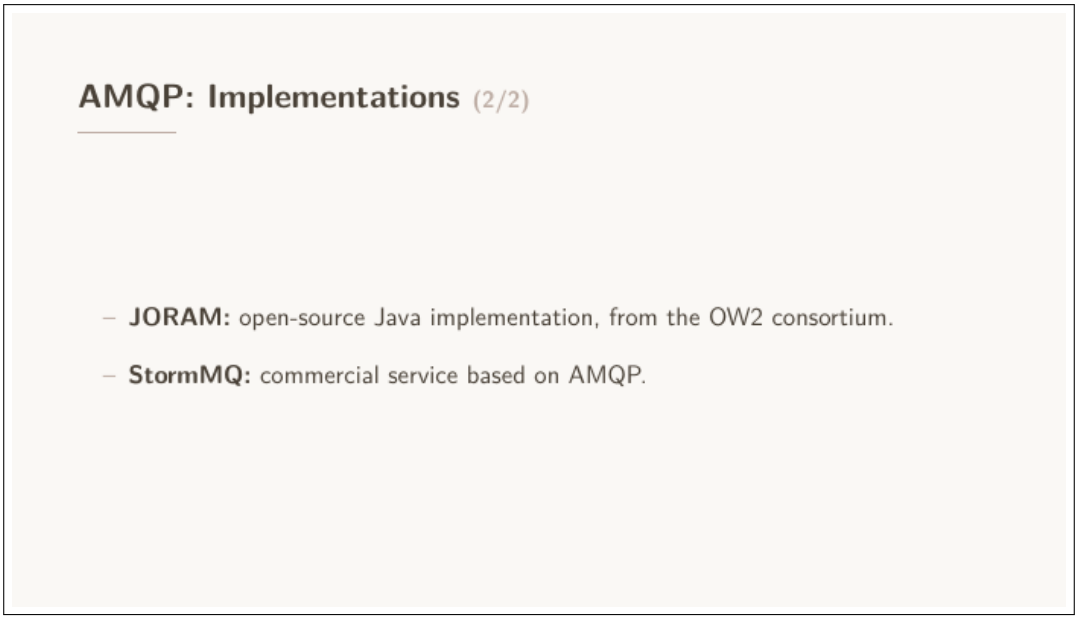
Tools and portability. The open standard allows independent traffic analysis tools, useful in diagnosis and audit, and brings the programming interfaces of implementations closer together, reducing migration effort.

Slide 22. AMQP: Implementations (1/2)



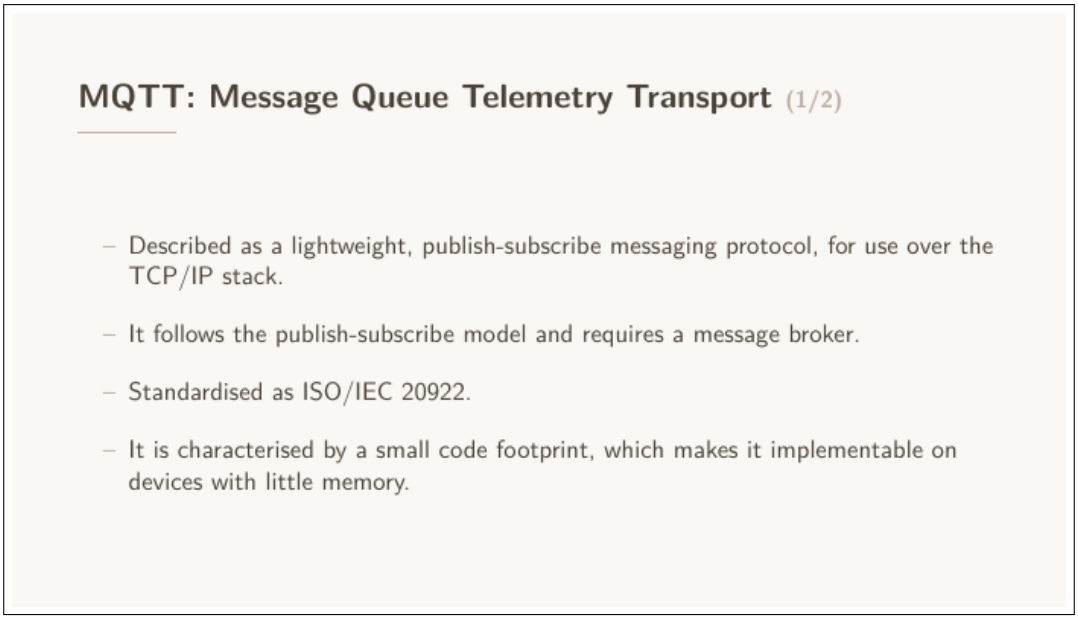
AMQP implementations. Qpid, ActiveMQ, RabbitMQ, Azure services and Solace. The student need not memorise the list, but should know that there are several implementations and that RabbitMQ is the one in the lab.

Slide 23. AMQP: Implementations (2/2)



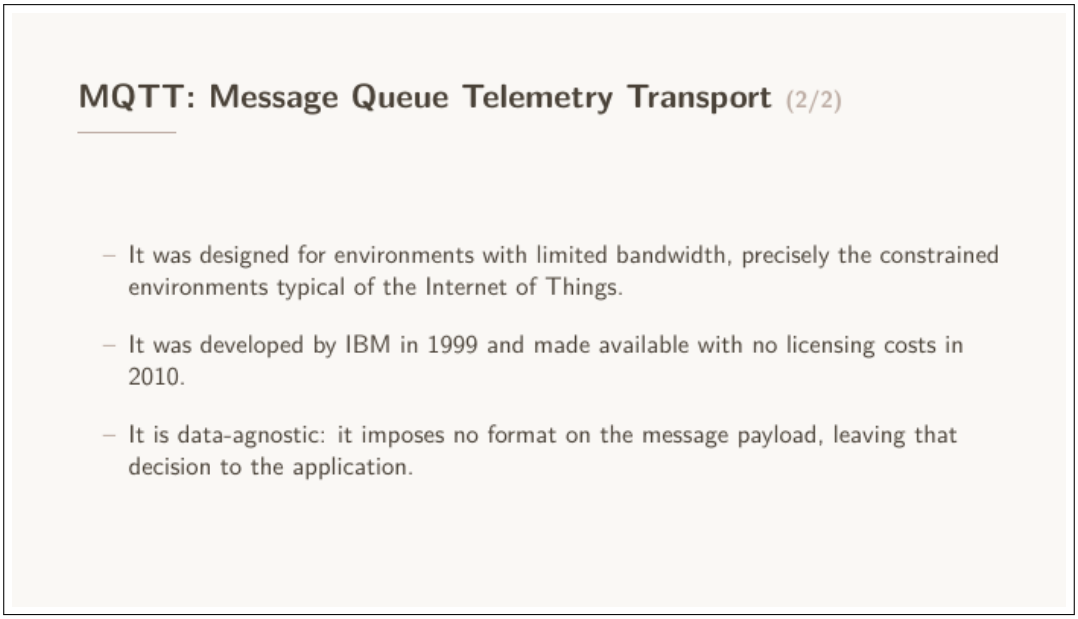
Other implementations. JORAM and StormMQ complete the picture. The point is variety: AMQP is not tied to a single vendor.

Slide 24. MQTT: Message Queue Telemetry Transport (1/2)



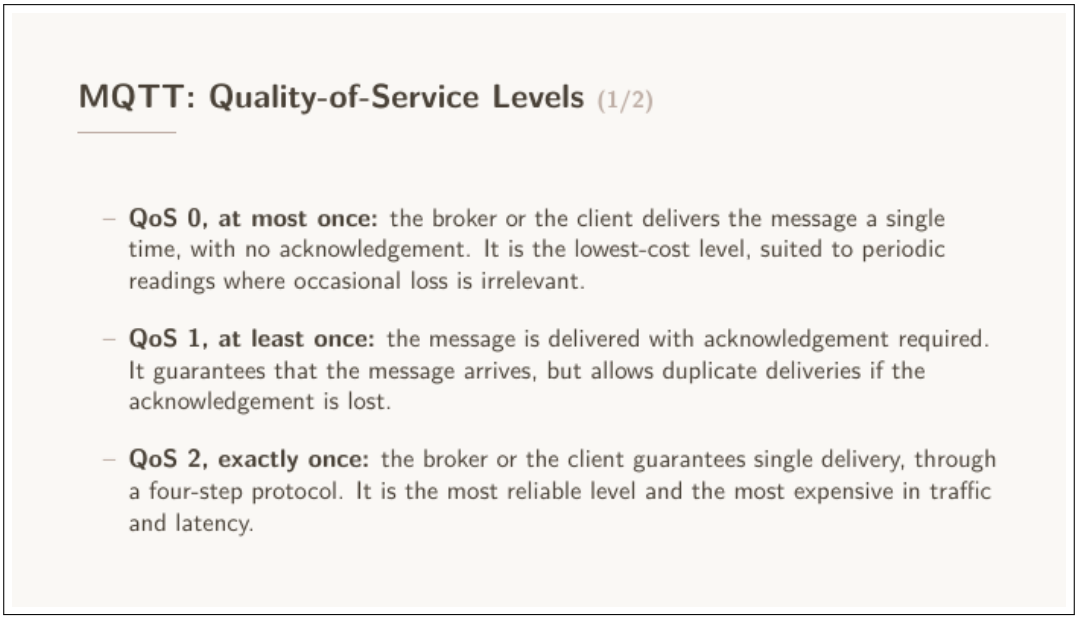
MQTT: light and standardised. MQTT is a lightweight publish-subscribe protocol, standardised as ISO/IEC 20922, with a small code footprint. It requires a broker.

Slide 25. MQTT: Message Queue Telemetry Transport (2/2)



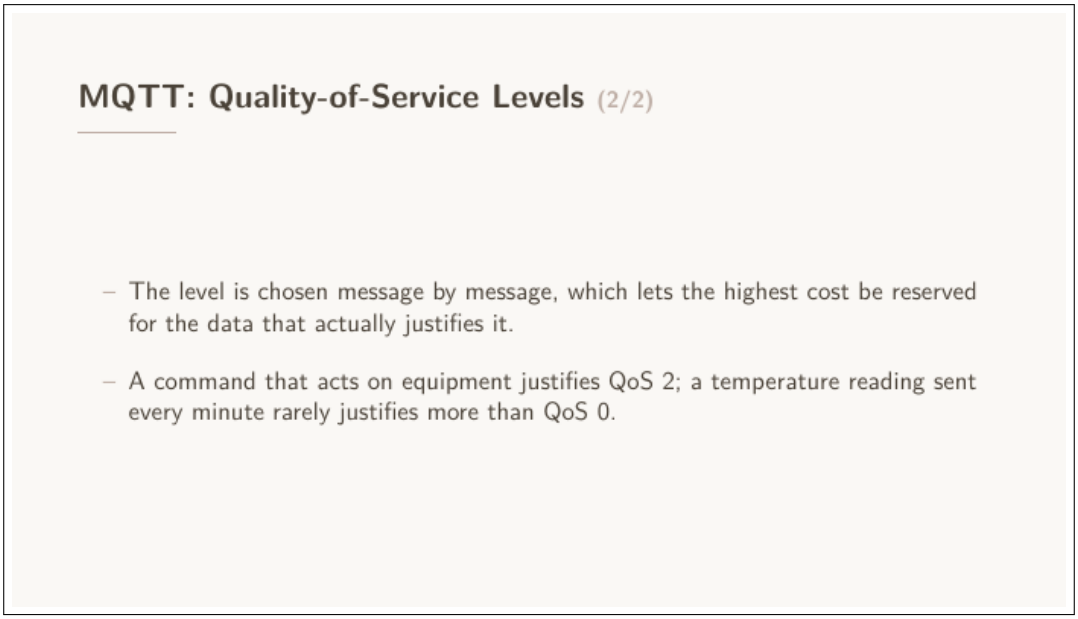
For constrained networks. It was designed for limited bandwidth, was born at IBM in 1999 and has been free since 2010. It is data-agnostic: the payload format is the application's decision.

Slide 26. MQTT: Quality-of-Service Levels (1/2)



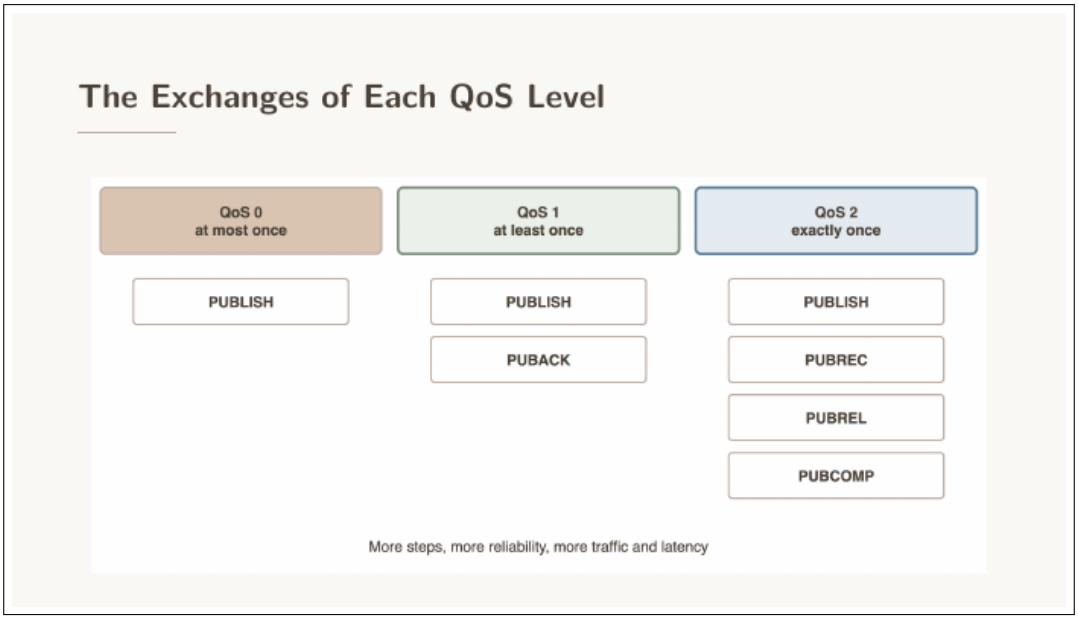
Three QoS levels. QoS 0 delivers at most once, QoS 1 at least once, QoS 2 exactly once. Each level adds exchanges, traffic and latency.

Slide 27. MQTT: Quality-of-Service Levels (2/2)



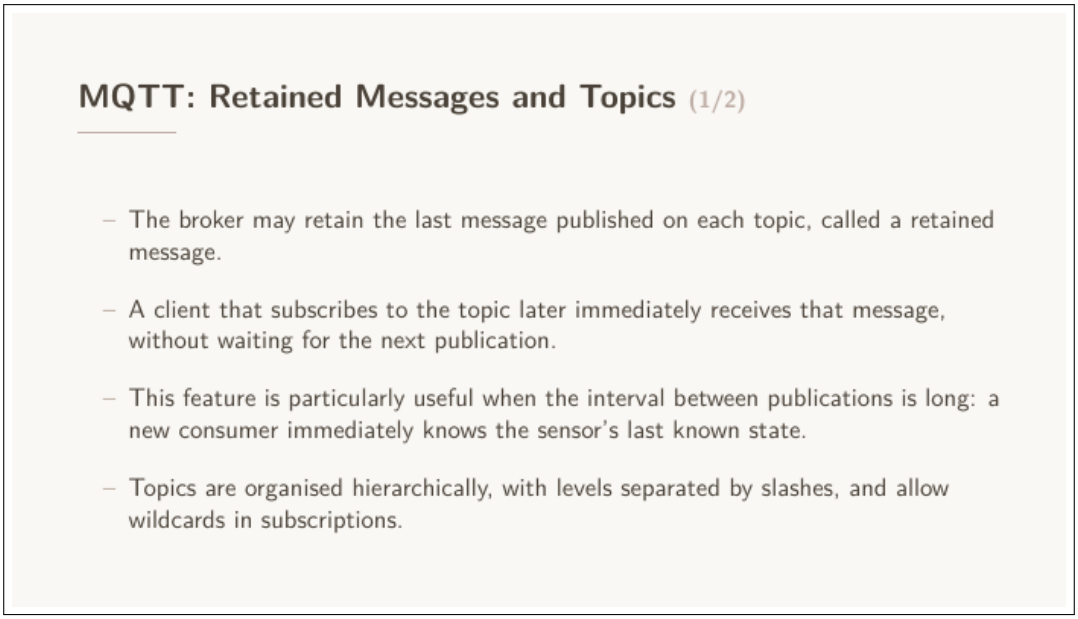
Choosing per message. The level is chosen message by message. A command to equipment justifies QoS 2, and a temperature reading every minute rarely justifies more than QoS 0.

Slide 28. The Exchanges of Each QoS Level



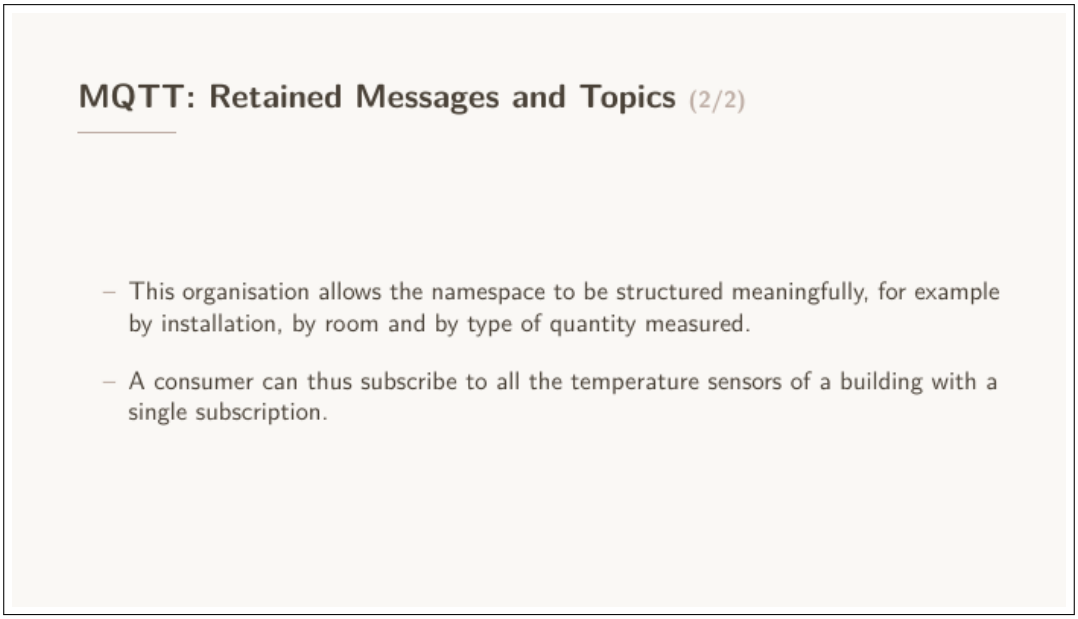
The exchanges of each level. QoS 0 is a single PUBLISH. QoS 1 adds a PUBACK. QoS 2 uses four messages: PUBLISH, PUBREC, PUBREL and PUBCOMP. Check: why can QoS 1 duplicate and QoS 2 not?

Slide 29. MQTT: Retained Messages and Topics (1/2)



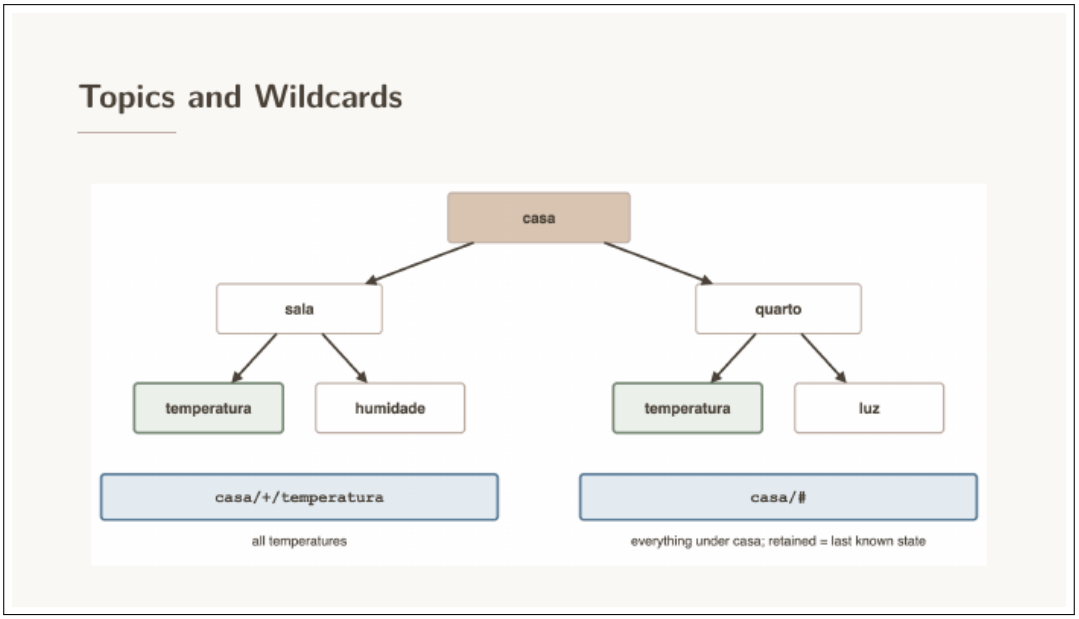
Retained messages. The broker stores the last message of each topic and delivers it to whoever subscribes later. When publications are spaced apart, a new consumer immediately knows the last state.

Slide 30. MQTT: Retained Messages and Topics (2/2)



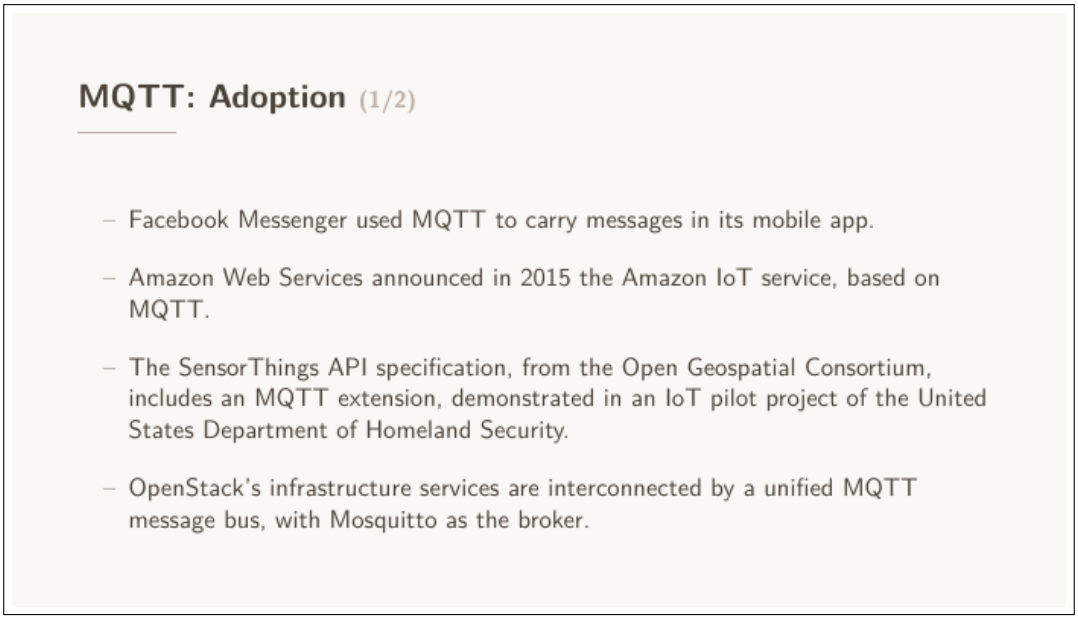
Topic hierarchy. Levels are separated by slashes and let the namespace be structured by installation, room and quantity. A single subscription catches all the temperature sensors of a building.

Slide 31. Topics and Wildcards



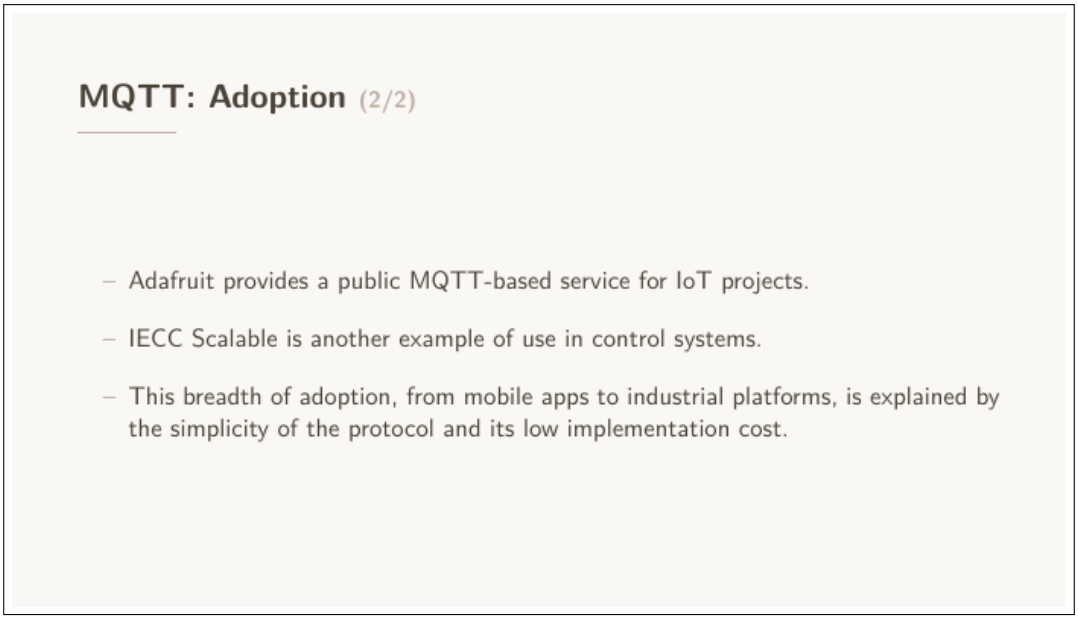
Wildcards. The plus sign replaces one level and the hash replaces all levels below. The figure shows `casa/+/temperatura` and `casa/#`. Check: which topics does each catch?

Slide 32. MQTT: Adoption (1/2)



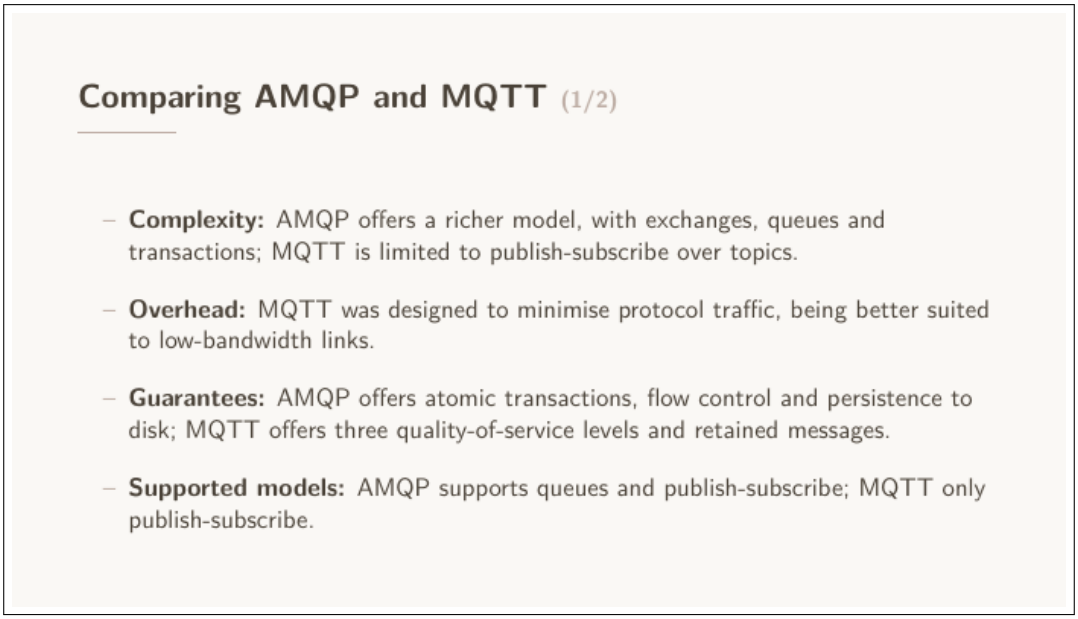
Who uses MQTT. Mobile messaging, cloud services, geospatial standards and infrastructures such as OpenStack use MQTT. They are examples of different contexts sharing the need for simplicity.

Slide 33. MQTT: Adoption (2/2)



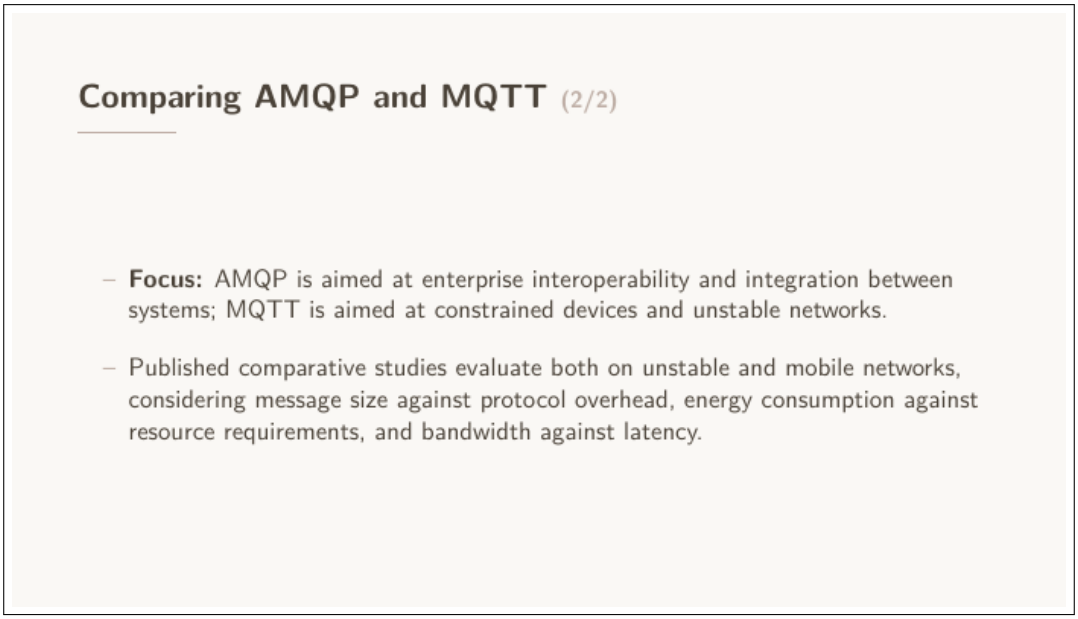
Simplicity as the reason. The breadth of adoption is explained by the simplicity of the protocol and its low implementation cost.

Slide 34. Comparing AMQP and MQTT (1/2)



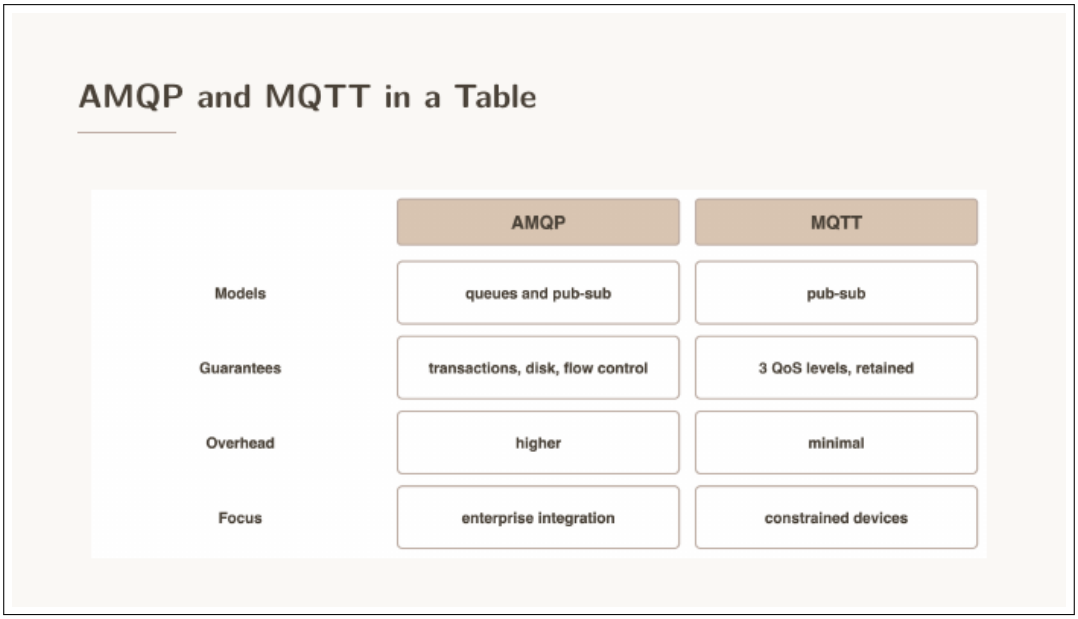
Complexity and overhead. AMQP is richer, with exchanges, queues and transactions. MQTT is limited to topic publish-subscribe and minimises protocol traffic.

Slide 35. Comparing AMQP and MQTT (2/2)



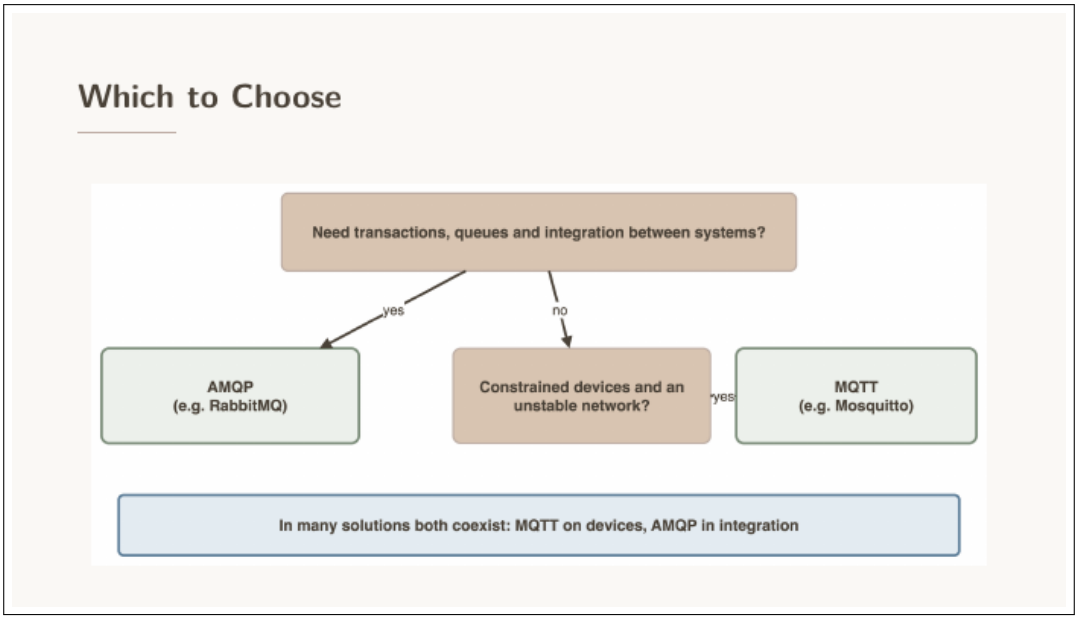
Guarantees and focus. AMQP offers transactions, flow control and persistence. MQTT offers three QoS levels and retained messages. One is aimed at enterprise integration and the other at constrained devices.

Slide 36. AMQP and MQTT in a Table



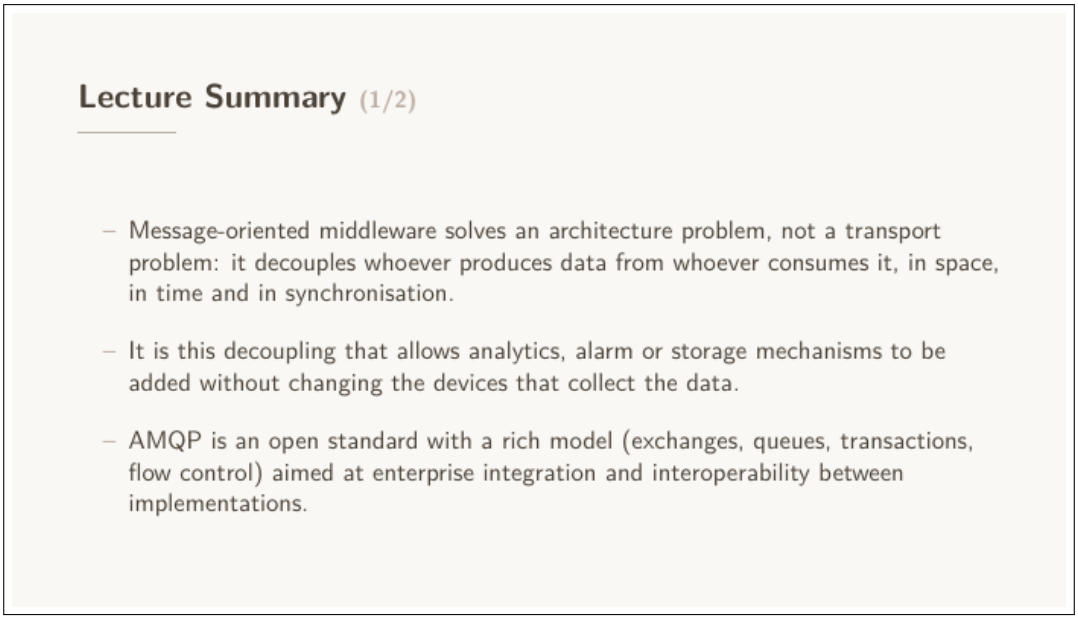
The comparison as a table. Models, guarantees, overhead and focus. You should be able to rebuild the table.

Slide 37. Which to Choose



Which to choose. The decision starts by asking whether transactions, queues and integration between systems are needed. If not, and the environment is constrained, choose MQTT. Many solutions use both.

Slide 38. Lecture Summary (1/2)



The essentials: decoupling. MOM solves an architecture problem, not a transport one. Decoupling lets analytics, alarms or storage be added without changing the devices.

Slide 39. Lecture Summary (2/2)



From sensor to middleware. AMQP is rich and open, MQTT is minimalist and constrained. Both use TCP and a broker. The next lecture connects a Zigbee network to an MQTT broker and closes the path.

Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. What three decouplings does a broker offer?
2. What is the difference between the queue model and the publish-subscribe model?
3. What is the role of the exchange in AMQP and which types do you know?
4. What distinguishes QoS 0, 1 and 2 in MQTT and when is each used?
5. Which topics do the subscriptions `casa/+/temperatura` and `casa/#` catch?
6. What criteria underlie the choice between AMQP and MQTT?

References

- OASIS. *MQTT Version 3.1.1.* OASIS Standard, 2014; ISO/IEC 20922:2016.
- OASIS. *Advanced Message Queuing Protocol (AMQP) Version 1.0.* OASIS Standard, 2012.
- VMware. *RabbitMQ Tutorials.* https://www.rabbitmq.com/tutorials.
- N. Naik. *Choice of Effective Messaging Protocols for IoT Systems: MQTT, CoAP, AMQP and HTTP.* IEEE ISSE, 2017.